

Protection

Definition

Protection means keeping computer information and system resources safe from unauthorized access, misuse, or damage. It is one of the important duties of the operating system.

1. Why Protection is Needed

Main Idea

Computers store a huge amount of important and confidential information. Users often want this information to remain private and secure.

Examples of Sensitive Information

Protected information may include:

- email,
- business plans,
- tax returns,
- personal files,
- official documents.

Role of the Operating System

It is the responsibility of the **operating system** to manage system security so that files and other resources are accessible only to authorized users.

2. File Protection in UNIX

UNIX Protection Code

In UNIX, each file is protected by a **9-bit binary protection code**. This code determines who can read, write, or execute the file.

Three 3-Bit Fields

The 9 bits are divided into three groups:

- one field for the **owner**,
- one field for the **group**,
- one field for **others**.

3. The rwx Bits

Meaning of rwx

Each 3-bit field contains:

- **r** = read permission
- **w** = write permission
- **x** = execute permission

These are called the **rwx bits**.

Permission Groups

So permissions are checked separately for:

- the file owner,
- members of the owner's group,
- all other users.

4. Example of UNIX Protection Code

Example

Consider the protection code:

`rwxr-x-x`

This means:

- **Owner:** `rw` → read, write, execute
- **Group:** `r-x` → read, execute only
- **Others:** `-x` → execute only

Meaning of Dash

A dash - means that the corresponding permission is **absent**.

5. Protection of Directories

Directory Permission

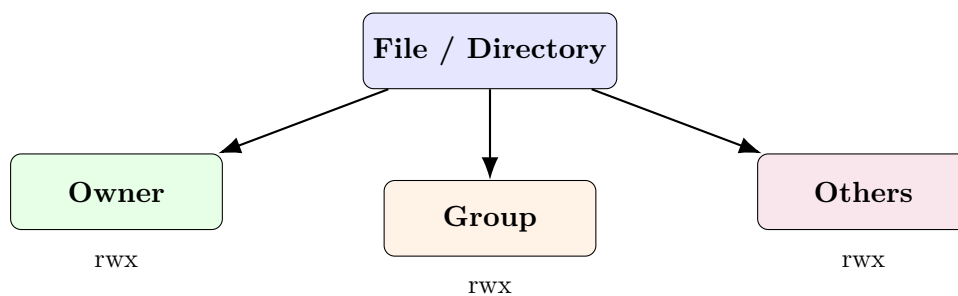
For a **directory**, the x bit means **search permission**. It allows a user to search or move through that directory.

Directory Permissions

Thus, directory permissions help control:

- who can view directory contents,
- who can modify files in it,
- who can search through it.

6. Simple Permission Diagram



7. Security Beyond File Protection

Broader Security

File protection is only one part of operating-system security. There are many other protection issues as well.

Other Security Threats

The operating system must also protect the system from:

- unwanted human intruders,
- malicious software,
- viruses,
- unauthorized access attempts.

8. Why OS Security Matters

Importance

Good protection is necessary to:

- keep user data confidential,
- prevent accidental damage,
- stop unauthorized access,
- keep the system reliable and safe.